

Sushan Adhikari

Computer Engineering Graduate | AI/ML | NLP | Computer Vision

sushan.adhikari2060@gmail.com | +977-9810538507 | Nepal | [Portfolio](#) | [Google Scholar](#) | [LinkedIn](#)

PROFILE

Computer Engineer (B.Eng. CGPA 3.90/4.0, Kathmandu University; exchange semester at IIT Palakkad) with research experience in NLP, Computer Vision, and LLMs. First-authored a demonstration paper at ACM ICAIL 2026 and two peer-reviewed journal publications; three further works in press or ongoing. Industry experience spans real-time CV inference pipelines and MLOps infrastructure. Seeking a fully-funded PhD with RA/TA support in AI/ML, NLP, or Computer Vision.

EDUCATION

B.Eng. Computer Engineering — Kathmandu University, School of Engineering Jan 2022 – Feb 2026

CGPA: 3.90/4.0 [\[Transcript\]](#)

Coursework: Data Structures & Algorithms, Computer Vision, Databases, Computer Architecture, Statistics & Probability, Differential Equations

Exchange Semester, B.Tech CSE — Indian Institute of Technology Palakkad

Jan – May 2025

Coursework: Introduction to AI, Linear Algebra, Compiler Design, Graph Theory & Combinatorics

RESEARCH & PUBLICATIONS

Conference

Directional Asymmetry in Low-Resource Legal Machine Translation: A Nepali-English Case Study *ACM ICAIL 2026, Jun 2026*

S. Adhikari, S. Sharma, D. Lamichhane, R. Chulyadyo, B. K. Bal

[\[GitHub\]](#) · [\[Acceptance Email\]](#)

- Curated the first publicly available Nepali-English legal parallel corpus (5,024 sentence pairs) and fine-tuned mBART-50 and NLLB-200 under unidirectional and bidirectional training regimes.
- Bidirectional training improves Nepali→English by ≈ 5 BLEU but degrades English→Nepali by 6.3 BLEU in mBART-50, attributing the asymmetry to Nepali's morphological complexity causing representational interference.

Journal

Enhancing Ethical Reasoning in Tiny LLMs via Fine-Tuning and Multi-Agent Consensus *KUSET Vol.18 No.2, Dec 2024*

S. Adhikari, S. Sharma, D. Lamichhane, S. Sigdel

[\[GitHub\]](#) · [\[NCCI Certificate\]](#)

- Fine-tuned TinyLlama-1.1B with LoRA on 1,000 synthetic ethical dilemmas; deployed three expert agents (utilitarian, deontological, virtue ethics) with a confidence-weighted consensus mechanism.
- Agents achieve 95+ % philosophical consistency (deontological: 97.8 %, utilitarian: 95.2 %, virtue: 96.5 %), validated via human Likert-scale evaluation.

Detecting Image Forgeries and Deepfakes: A Comparative Study of CNN and Transformer Models *KUSET Vol.18 No.2, Dec 2024*

P. Kafle, S. Adhikari, A. M. Shakya, N. Ghimire, G. Sharma

- Benchmarked InceptionV3 and ViT on a 140,000+ image dataset (photoshopping, splicing, copy-move, StyleGAN/deepfakes); InceptionV3 outperforms ViT by 5.51 % accuracy and 10.8 % recall, demonstrating CNN advantage where local texture cues are discriminative.

Physics-Informed Data Augmentation for Sediment Concentration Prediction in Himalayan Rivers *InJET-InDev Vol.2 No.2, Feb 2026*

U. Adhikari, M. R. Kafle, S. Adhikari

[\[LEC Certificate\]](#)

- Physics-informed augmentation outperforms 5 classical baselines for sediment prediction in data-scarce Himalayan rivers: $R^2 +5.5$ %, RMSE -24.3 %, MAE -47.6 % (5-fold CV).

In Press / Ongoing

RAG for Theoretical CS Education: Algorithm Analysis & Complexity Theory

IOP Conf. Series (USC 2025), In Press

S. Adhikari, S. Sharma, D. Lamichhane, U. Adhikari [\[GitHub\]](#) · [\[USC Certificate\]](#)

- Domain-specialized RAG over 800+ lecture slides and 150+ proof templates; 100 % response success on 179 exam-style questions, pedagogical quality score 0.7620.

Automated Mathematical Animation Generation: Dataset Requirements and Validation Gaps

Ongoing (Target: Elsevier)

[\[GitHub\]](#)

- Fine-tuned DeepSeek-Coder-1.3B on 1,500-sample Manim dataset; exposed a validation gap where training loss converges yet test-set execution success is 0 %, identifying dataset scale as the core bottleneck.

Explainability Methods for Hybrid CNN-Transformer Document Layout Analysis

Ongoing [\[GitHub\]](#)

- Compared Grad-CAM, Attention Rollout, and LIME on PubLayNet/DocLayNet; CNN branch captures local boundary cues, Transformer captures global page structure.

TECHNICAL SKILLS

Languages & ML:	Python; PyTorch, Scikit-Learn, OpenCV, YOLOv8; Pandas, NumPy
Inference & CV:	NVIDIA DeepStream, Triton Inference Server, GStreamer, TensorRT
MLOps & DevSecOps:	Docker, AWS/EC2, Terraform, CI/CD (SAST/DAST, DefectDojo, CodeRabbit), Git
Backend & Data:	FastAPI, Flask, REST APIs, PostgreSQL, Spring Boot (integration)
Other:	Monorepo architecture, CODEOWNERS management, $\LaTeX$ /Overleaf

INTERNSHIP EXPERIENCE

- **Junior MLOps Developer** — KyraWorks Jan – Apr 2026 [\[Certificate\]](#)
Promoted from CV Intern; set up company-wide monorepo with CODEOWNERS governance; integrated SAST/DAST pipelines using DefectDojo and CodeRabbit; containerized services with Docker and managed deployments on AWS EC2 via Terraform.
- **Computer Vision Intern** — KyraWorks Oct 2025 – Jan 2026 [\[Certificate\]](#)
Built and optimized real-time inference pipelines using NVIDIA DeepStream and Triton Inference Server; benchmarked multi-camera (10+ streams) throughput for production deployment.
- **Full-Stack Development Intern** — National Pension Management System, Nepal Jun – Oct 2025 [\[Certificate\]](#)
Contributed to national pension platform serving 300k+ users; implemented liveness detection (Amazon Rekognition) for anti-spoofing; assisted in DB schema design (Spring Boot, PostgreSQL).
- **AI Data Volunteer** — [Mercuri.world](#) (*Global Non-profit*) Jan 2025 – Present
Sole Nepali member (150+ global volunteers); contributed to job taxonomy design and pipeline planning for an employment platform targeting individuals with mental illness.

PROJECTS & STARTUPS

- **Nurvexa** — Co-founder [nurvexalabs.com](#) 2025 – Present
RAG-powered nursing exam preparation platform (launching); adaptive question generation and personalized study plans targeting Nepal's PCL Nursing licensing exam.
- **PrepGraduate** — Co-founder [prepgraduate.com](#) 2026 – Present
AI-driven GRE preparation platform (in development); full-length adaptive mock tests and score analytics for graduate program applicants.

ACHIEVEMENTS & LEADERSHIP

- **Hult Prize OnCampus Winner** (2024) — Led computer vision component for fish-disease detection using a Roboflow pretrained model (~60% baseline accuracy) as proof-of-concept; represented Nepal at Regional Summit, Bangkok, Thailand.
- **Finalist** — ICT Awards 2024, Rising Star Innovation Category.
- **Winner, Health Track** — AI Crusade 2023 (first AI hackathon in Nepal). [\[Certificate\]](#)
- **Scholarships:** Mahatma Gandhi Scholarship (Indian Embassy, 2019–2021); Merit-based scholarship, Kathmandu University (1st Semester, 2022).
- **Internal Relations & Ops Lead** — KU HackFest 2024: managed logistics, judging, and volunteers for 48-hour hackathon (~200 team applications).

CERTIFICATIONS

[Machine Learning Specialization](#) — Andrew Ng (Stanford/Coursera, 2026) · [Embeddings for NLP](#) (OpenHPI, 2025)

REFERENCES

Prof. Dr. Bal Krishna Bal
Assoc. Dean, School of Eng., KU
bal@ku.edu.np
+977-11415100

Prof. Dr. Gajendra Sharma
Professor, Dept. of CSE, KU
gajendra.sharma@ku.edu.np
+977-9840093887

Dr. Rajani Chulyadyo
Asst. Professor, Dept. of CSE, KU
rajani.chulyadyo@ku.edu.np
+977-9803293409